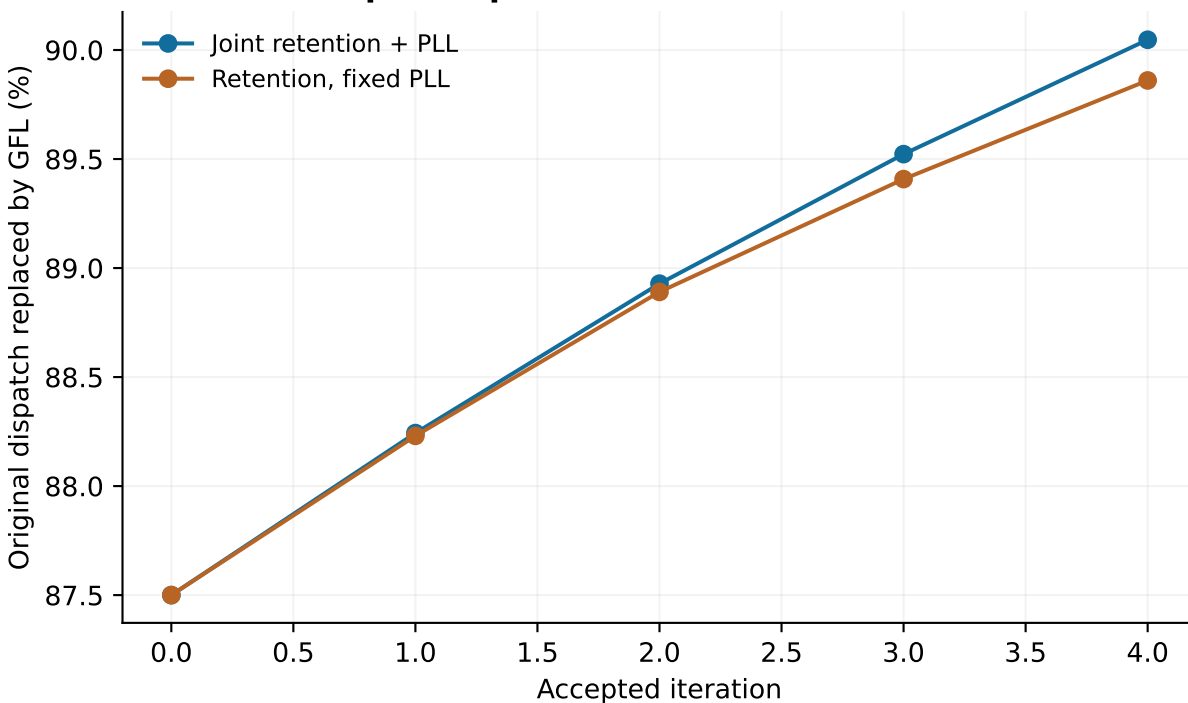
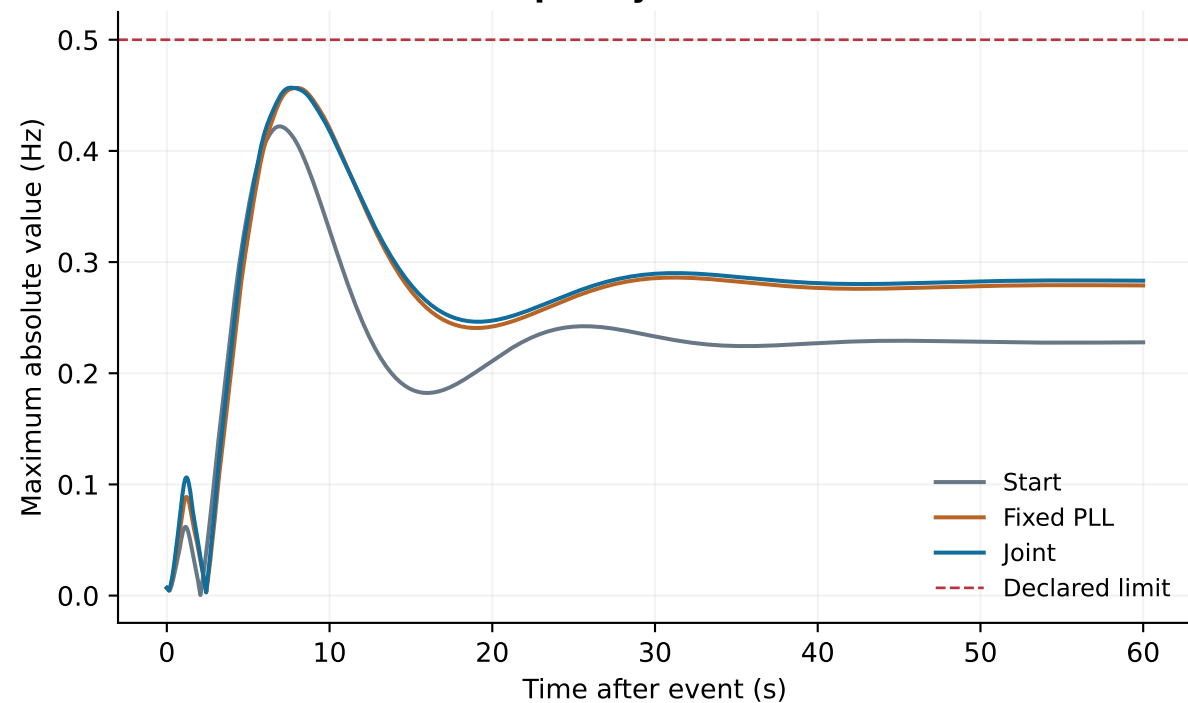


Analytical iterative co-design: a bounded IEEE-39 experiment

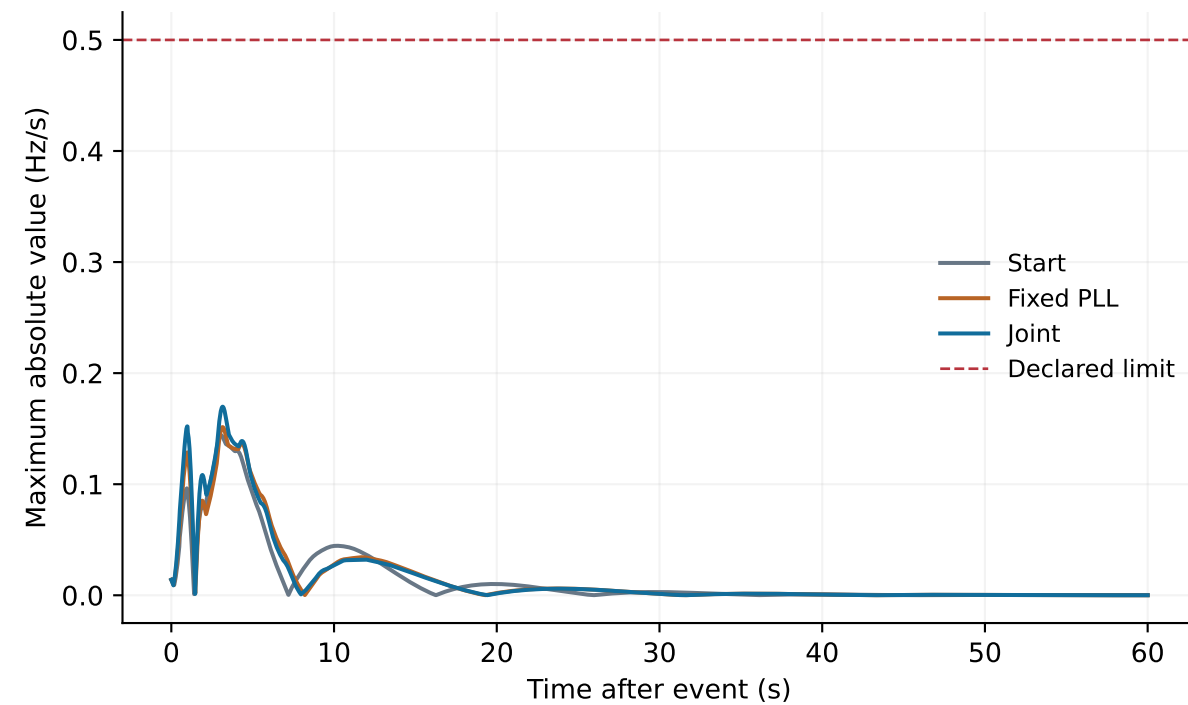
A Explicit updates on the nonlinear model



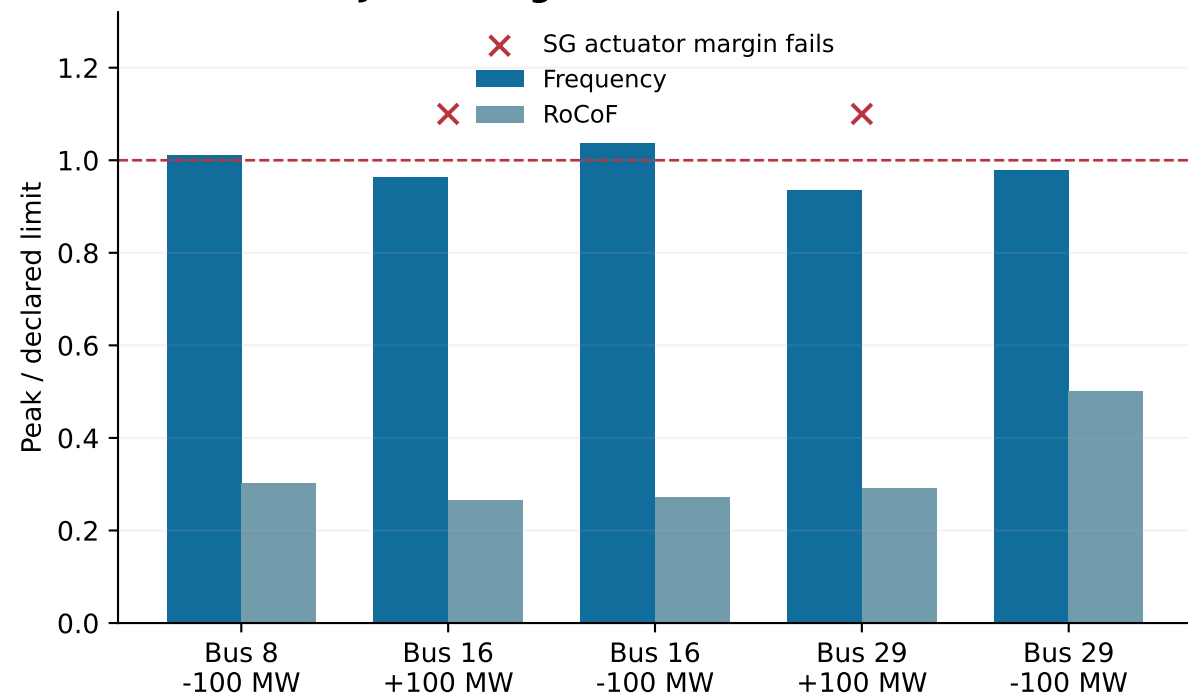
B Frequency: all 39 buses



C RoCoF: causal 0.5 s window



D Joint design: external-event checks



Design event: +100 MW ZIP setpoint at bus 8 | Physical DC convention | 60 s validation | No global optimality claim